Adhoc Sensor Networks UNIT1

ad hoc and sensor network (Jawaharlal Nehru Technological University Hyderabad College of Engineering Sultanpur)

UNIT-I

Introduction to AdHoc Networks-Characteristics of MANETs, Applications of MANETs and Challenges of MANETs.

Routing in MANETs - Criteria for classification, Taxonomy of MANET routing algorithms, Topology-based routing algorithms-**Proactive**: DSDV; **Reactive**: DSR, AODV; Hybrid: ZRP; Position-basedroutingalgorithms-**LocationServices**-DREAM, Quorum-based; **ForwardingStrategies:GreedyPacket,Restricted** Directional Flooding-DREAM,LAR

Introduction

Simply stating, a Mobile Ad hoc NETwork (MANET) is one that comes together as needed, not necessarily with any support from the existing Internet infrastructure or any other kind of fixed stations. We can formalize this statement by defining an ad hoc network as an autonomous system of mobile hosts (also serving as routers) connected by wireless links, the union of which forms a communication network modeled in the form of an arbitrary graph. This is in contrast to the well-known single hop cellular network model that supports the needs of wireless communication by installing base stations as access points. In these cellular networks, communications between two mobile nodes completely rely on the wired backbone and the fixed base stations. In a MANET, no such infrastructure exists and the network topology may dynamically change in an unpredictable manner since nodes are free to move.

As for the mode of operation, ad hoc networks are basically peer-to-peer multi- hop mobile wireless networks where information packets are transmitted in a store-and- forward manner from a source to an arbitrary destination, via intermediate nodes as shown in Figure 1. As the nodes move, the resulting change in network topology must be made known to the other nodes so that outdated topology information can be updated or removed. For example, as MH2 in Figure 1 changes its point of attachment from MH3 to MH4 other nodes part of the network should use this new route to forward packets to MH2.

Note that in Figure 1, and throughout this text, we assume that it is not possible to have all nodes within range of each other. In case all nodes are close-by within radio range, there are no routing issues to be addressed. In real situations, the power needed toobtain complete connectivity may be, at least, infeasible, not to mention issues such as battery life. Therefore, we are interested in scenarios where only few nodes are within radio range of each other.

Figure 1 raises another issue of symmetric (bi-directional) and asymmetric (unidirectional) links. As we shall see later on, some of the protocols we discuss consider symmetric links with associative radio range, i.e., if (in Figure 1) MH1 is within radio range of MH3, then MH3 is also within radio range of MH1. This is to say that the communication links are symmetric. Although this assumption is not always valid, it is usually made because routing in asymmetric networks is a relatively hard task [Prakash 1999]. In certain cases, it is possible to find routes that could avoid asymmetriclinks, since it is quite likely that these links imminently fail. Unless stated otherwise, throughout this text we consider symmetric links, with all nodes having identical capabilities and responsibilities.

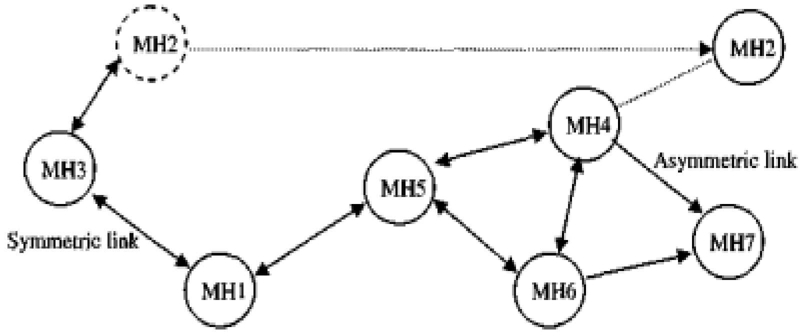

Figure 1.1 – A mobile ad hoc network (MANET)

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Table 1.1 – Important characteristics of a MANET

Characteristic	Description
Dynamic Topologies	Nodes are free to move arbitrarily with different speeds; thus, the network topology may change randomly and at unpredictable times.
Energy-constrained Operation	Some or all of the nodes in an ad hoc network may rely on batteries or other exhaustible means for their energy. For these nodes, the most important system design optimization criteria may be energy conservation.
Limited Bandwidth	Wireless links continue to have significantly lower capacity than infrastructured networks. In addition, the realized throughput of wireless communications – after accounting for the effects of multiple access, fading, noise, and interference conditions, etc., is often much less than a radio's maximum transmission rate.
Security Threats	Mobile wireless networks are generally more prone to physical security threats than fixed-cable nets. The increased possibility of eavesdropping, spoofing, and minimization of denial-of-service type attacks should be carefully considered.

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Wireless Sensor Networks [Estrin 1999, Kahn 1999] is an emerging application area for ad hoc networks which has been receiving a large attention. The idea is that a collection of cheap to manufacture, stationary, tiny sensors would be able to sense, coordinate activities and transmit some physical characteristics about the surrounding environment to an associated base station. Once placed in a given environment, these sensors remain stationary. Furthermore, it is expected that power will be a major drivingissue behind protocols tailored to these networks, since the lifetime of the battery usually defines the sensor's lifetime. One of the most cited examples is the battlefield urveillance of enemy's territory wherein a large number of sensors are dropped from an airplane so that activities on the ground could be detected and communicated. Other potential commercial fields include machinery prognosis, bio sensing and environmentalmonitoring.

This rest of this text is organized as follows. We initially provide necessary background on ad hoc networking by illustrating its diverse applications. Next, we cover the routing aspect in a MANET, considering both unicast and multicast communication. MAC issues related to a MANET are then illustrated. Following, sensor networks, its diverse applications, and associated routing protocols are discussed. Finally, we conclude this text by discussing the current standard activities at both IETF and the Bluetooth SIG, and also bringing up some open problems that have not received much attention so far and still need to be addressed.

Characteristics of MANET

Let's have a look at some of the characteristics of MANET:

1. **Dynamic topologies**:- As nodes are free to move in any direction, the network topology can alter at any time and is mainly composed of bidirectional links. A unidirectional link may sometimes exist when the transmission power of two nodes differs.

2. **Bandwidth-constrained and variable capacity links**:- Wireless links continue to have much lesser capacity than infrastructure networks.
3. **Energy-constrained operation**:- Batteries or other non-renewable sources may power some or all of the nodes in a MANET. Energy conservation may be the most essential system design optimization requirement for these nodes or devices.
4. **Limited physical security**:- MANETs are generally more vulnerable to physical security threats than wireline networks. The increased risk of eavesdropping, spoofing, and denial of service (DoS) attacks should be carefully examined. Many existing link security solutions are frequently used within wireless networks to reduce security concerns.
5. **Less Human Intervention**:- They require minimal human involvement to configure the network, hence they are dynamically autonomous

Advantages of MANETs

1. Each node can act as both a router and a host, demonstrating its autonomous nature.
2. Separation from the central network administration.
3. Highly expandable and suitable for the addition of new network hubs.
4. Nodes that self-configure and self-heal do not require human involvement.
5. In MANETs infrastructure is not required because it is a decentralized network.
6. Due to the multi-hop approach in which information is conveyed, decentralized networks are often more robust than centralized networks. In a cellular network, for example, if a base station fails, coverage is lost; however, the likelihood of a single point of failure in a MANET is greatly decreased because data can travel via several paths.
7. Other advantages of MANETs over fixed-topology networks include flexibility (mobile devices can be used to form an ad hoc network anywhere), scalability (you can quickly add more nodes to the network), and cheaper management expenses (no need to build infrastructure first).

Disadvantages of MANETs

- There are no authorization facilities.

- Resources are limited due to various constraints such as noise, interference situations, and so on.
- High latency means data is transferred between two sleeping nodes with a significant delay.
- Due to inadequate physical security, they are more vulnerable to attacks.

Applications of MANETs

There are many applications to ad hoc networks. As a matter of fact, any day- to-day application such as electronic email and file transfer can be considered to be easily deployable within an ad hoc network environment. Web services are also possible in case any node in the network can serve as a gateway to the outside world. In this discussion, we need not emphasize the wide range of military applications possible with ad hoc networks. Not to mention, the technology was initially developed keeping in mind the military applications, such as battlefield in an unknown territory where an infrastructured network is almost impossible to have or maintain. In such situations, the ad hoc networks having self-organizing capability can be effectively used where other technologies either fail or cannot be deployed effectively. Advanced features of wireless mobile systems, including data rates compatible with multimedia applications, global roaming capability, and coordination with other network structures, are enabling new applications. Some well-known ad hoc network

Applications are:

- **Collaborative Work** – For some business environments, the need for collaborativecomputing might be more important outside office environments than inside. After all, it is often the case where people do need to have outside meetings to cooperateand exchange information on a given project.
- **Crisis-management Applications** – These arise, for example, as a result of natural disasters where the entire communications infrastructure is in disarray. Restoring communications quickly is essential. By using ad hoc networks, an infrastructure could be set up in hours instead of days/weeks required for wire-line communications.
- **Personal Area Networking and Bluetooth** – A personal area network (PAN) is a short-range, localized network where nodes are usually associated with a given

Person. These nodes could be attached to someone's pulse watch, belt, and so on.

In these scenarios, mobility is only a major consideration when interaction among several PANs is necessary, illustrating the case where, for instance, people meet in real life. Bluetooth [Haarsten 1998], is a technology aimed at, among other things, supporting PANs by eliminating the need of wires between devices such as printers, PDAs, notebook computers, digital cameras, and so on, and is discussed later.

Education via the internet: - Educational opportunities are available on the internet or in remote places due to the practical impossibility of providing pricey last-mile wireline internet access to all users in these areas.

Virtual Navigation:- A big metropolis' physical properties, including its buildings, streets, and other physical features, are graphically represented in a remote database. Additionally, they might be able to "virtually" view a building's interior layout, including a strategy for an emergency evacuation, or locate potential places of interest.

Vehicular area network(VANETs):- This is a growing and precious ad-hoc network application for providing emergency services and other information. This works equally well in both urban and rural settings. The fundamental and necessary data interchange in a specific circumstance.

Challenges of MANET

- **Limited Bandwidth**
  The wireless networks have a limited bandwidth in comparison to the wired networks. Wireless link has lower capacity as compare to infrastructure networks. The effect of fading, multiple accesses, interference condition is very low in ADHOC networks in comparison to maximum radio transmission rate.
- **Dynamic topology**
  Due to dynamic topology the nodes has less truest between them. I some settlement are found between the nodes then it also make trust level questionable.
- **High Routing**

In ADHOC networks due to dynamic topology some nodes changes their position which affects the routing table.

- **Problem of Hidden terminal**

  The Collision of the packets are held due to the transmission of packets by those node which are not in the direct transmission range of sender side but are in range of receiver side.

- **Transmission error and packet loss**

  By increasing in collisions , hidden terminals, interference, uni-directional links and by the mobility of nodes frequent path breaks a higher packet loss has been faced by ADHOC networks.

- **Mobility**

  Due to the dynamic behavior and changes in the network topology by the movement of the nodes .ADHOC networks faces path breaks and it also changes in the route frequently.

- **Security threats**

  New security challenges bring by Adhoc networks due to its wireless nature. In Adhoc networks or wireless networks the trust management between the nodes leads to the numerous security attacks.

- **Some Other Major Challenges in MANET**

  Some other challenges of the MANET are describe briefly as below:

- **Dynamic topologies**

  In Dynamic Topology the nodes are free to move in any direction. Topology of network changes rapidly and unpredictably by the time. Due to this topology the bidirectional and unidirectional routing exists. Challenging task is to transferring the packets between the nodes because the topologies are changes continuously.

- **Multicast Routing**

  Another challenge of MANET is multicast. The multicast dynamic this networks because the nodes are randomly changes its position. The nodes have multiple hopes instead of single hop and they are complex. The new device adds in the network need to know all the other nodes. To facilitate automatic optimal route selection dynamic update is necessary due to existence of node

- **Variability in capacity links due to Bandwidth Constraint**

  Being a link of wireless they continue with low capacity in comparison to the hardwired.
- **Power-constrained and operation**

  This is also considered as a challenge for the MANET network. The MANET is a network where all nodes rely on the batteries or some exhaustible source of energy. Conversion in the energy is the optimized criteria and an important system design. Lean power consumption is also used for the light weight mobile terminals. Conservation of power and power-aware routing is another aspect which must be considered.
- **Security and Reliability**

  Nasty Neighbor relaying packets is also a security problem alongwith other vulnerabilities connected. Different schemes are used for authentication and key management in distributed operations. Reliability problem is also a wireless link characteristic due to limited wireless transmission. Due to the broadcast of wireless medium packets loss and errors in the data occur. In comparison to the wired network the wireless networks are more vulnerable to security threats.
- **Quality of Service**
- In MANET the environment will change constantly so that it provides different quality of service levels which are challengeable. The random nature in the quality communication of MANET it is difficult to server good guarantee of service of the device. To support multimedia services adaptive Quality of Services can be implement over traditional resources.
- **Inter-networking**
- Communication with fixed networks is also expected from MANET in many cases. The existence of routing protocols in both the networks is quite challengeable for pleasant mobility management.

ROUTING IN A MANET

It has become clear that routing in a MANET is intrinsically different from traditional routing found on infra structured networks. Routing in a MANET depends on many factors including topology, selection of routers, and initiation of request, and specific underlying characteristic that could serve as a heuristic in finding the path quickly and efficiently. The low resource availability in these networks demands efficient utilization and hence the motivation for optimal routing in ad hoc networks. Also, the highly dynamic nature of these networks imposes severe restrictions on routing protocols specifically designed for them, thus motivating the study of protocols which aim at achieving routing stability.

One of the major challenges in designing a routing protocol [Jubin 1987] for ad hoc networks stems from the fact that, on one hand, a node needs to know at least the reachability information to its neighbors for determining a packet route and, on the other hand, the network topology can change quite often in an ad hoc network. Furthermore, as the number of network nodes can be large, finding route to the destinations also

requires large and frequent exchange of routing control information among the nodes. Thus, the amount of update traffic can be quite high, and it is even higher when high mobility nodes are present. High mobility nodes can impact route maintenance overhead of routing protocols in such a way that no bandwidth might remain leftover for the transmission of data packets [Corson 1996].

Proactive and Reactive Routing Protocols

Ad hoc routing protocols can be broadly classified as being Proactive (or table- driven) or Reactive (on-demand). Proactive protocols mandates that nodes in a MANET should keep track of routes to all possible destinations so that when a packet needs to beforwarded, the route is already known and can be immediately used. On the other hand, reactive protocols employ a lazy approach whereby nodes only discover routes to destinations on demand, i.e., a node does not need a route to a destination until that destination is to be the sink of data packets sent by the node.

Proactive protocols have the advantage that a node experiences minimal delay whenever a route is needed as a route is immediately selected from the routing table. However, proactive protocols may not always be appropriate as they continuously use a substantial fraction of the network capacity to maintain the routing information current. To cope up with this shortcoming, reactive protocols adopt the inverse approach by finding a route to a destination only when needed. Reactive protocols often consume much less bandwidth than proactive protocols, but the delay to determine a route can be significantly high and they will typically experience a long delay for discovering a route to a destination prior to the actual communication. In brief, we can conclude that no protocol is suited for all possible environments, while some proposals using a hybrid approach have been suggested.

Unicast Routing Protocols

Proactive Routing Approach

In this section, we consider some of the important proactive routing protocols.

Destination-Sequenced Distance-Vector Protocol

The destination-sequenced distance-vector (DSDV) [Perkins 1994] is a proactive hop-by-hop distance vector routing protocol, requiring each node to periodically broadcast routing updates. Here, every mobile node in the network maintains a routing table for all possible destinations within the network and the number of hops to each destination. Each entry is marked with a sequence number assigned by the destination node. The sequence numbers enable the mobile nodes to distinguish stale routes from new ones, thereby avoiding the formation of routing loops. Routing table updates are periodically transmitted throughout the network in order to maintain consistency in the table.

To alleviate the potentially large amount of network update traffic, route updates can employ two possible types of packets: full dumps or small increment packets. A full dump type of packet carries all available routing information and can require multiple network protocol data units (NPDUs). These packets are transmitted infrequently during periods of occasional movement. Smaller incremental packets are used to relay only the information that has changed since the last full dump. Each of these broadcasts should fit into a standard-size NPDU, thereby decreasing the amount of traffic generated. The mobile nodes maintain an additional table where they store the data sent in the incremental routing information packets. New route broadcasts contain the address of the destination, the number of hops to reach the destination, the sequence number of the information received regarding the destination, as well as a new sequence number unique to the broadcast. The route labeled with the most recent sequence number is always used. In the event that two updates have the same sequence number, the route with the smaller metric is used in order to optimize (shorten) the path. Mobiles also keep track of settling time of the routes, or the weighted average time that routes to a destination could fluctuate before the route with the best metric is received. By delaying the broadcast of a routing update by the length of the settling time, mobiles can reduce network traffic and optimize routes by eliminating those broadcasts that would occur if a better route could be discovered in the very near future.

Note that each node in the network advertises a monotonically increasing sequence number for itself. The consequence of doing it so is that when a node B decides that its route to a destination D is broken, it advertises the route to D with an infinite metric and a sequence number one greater than its sequence number for the route that has broken (making an odd sequence number). This causes any node A routing packets through B to incorporate the infinite-metric route into its routing table until node A hears a route to D with a higher sequence number.

The Wireless Routing Protocol

The Wireless Routing Protocol (WRP) [Murthy 1996] described is a table-based protocol with the goal of maintaining routing information among all nodes in the network. Each node in the network is responsible for maintaining four tables: Distance table, Routing table, Link-cost table, and the Message Retransmission List (MRL) table. Each entry of the MRL contains the sequence number of the update message, a re-transmission counter, an acknowledgment-required flag vector with one entry per neighbor, and a list of updates sent in the update message. The MRL records which updates in an update message need to be retransmitted and neighbors should acknowledge the retransmission.

Mobiles inform each other of link changes through the use of update messages. An update message is sent only between neighboring nodes and contains a list of updates (the destination, the distance to the destination, and the predecessor of the destination), as well as a list of responses indicating which mobiles should acknowledge (ACK) the update. After processing updates from neighbors or detecting a change in a link, mobiles send update messages to a neighbor. In the event of the loss of a link between two nodes, the nodes send update messages to their neighbors. The neighbors then modify their distance table entries and check for new possible paths through other nodes. Any new paths are relayed back to the original nodes so that they can update their tables accordingly.

Nodes learn about the existence of their neighbors from the receipt of acknowledgments and other messages. If a node is not sending messages, it must send a hello message within a specified time period to ensure connectivity. Otherwise, the lack of messages from the node indicates the failure of that link; this may cause a false alarm. When a mobile receives a hello message from a new node, that new node is added to the mobile's routing table, and the mobile sends the new node a copy of its routing table information.Part of the novelty of WRP stems from the way in which it achieves freedom from loops. In WRP, routing nodes communicate the distance and second-to-last hop

information for each destination in the wireless networks. WRP belongs to the class of path-finding algorithms with an important exception. It avoids the "count-to-infinity" problem by forcing each node to perform consistency checks of predecessor information reported by all its neighbors. This ultimately (although not instantaneously) eliminates looping situations and provides faster route convergence when a link failure occurs.

Reactive Routing Approach

In this section, we describe some of the most cited reactive routing protocols.

Dynamic Source Routing

The Dynamic Source Routing (DSR) [Johnson 1996] algorithm is an innovative approach to routing in a MANET in which nodes communicate along paths stored in source routes carried by the data packets. It is referred as one of the purest examples of an on-demand protocol [Perkins 2001]. In DSR, mobile nodes are required to maintain route caches that contain the source routes of which the mobile is aware. Entries in the route cache are continually updated as new routes are learned. The protocol consists of two major phases: route discovery and route maintenance. When a mobile node has a packet to send to some destination, it first consults its route cache to determine whether it already has a route to the destination. If it has an unexpired route to the destination, it will use this route to send the packet. On the other hand, if the node does not have such a route, it initiates route discovery by broadcasting a route request packet. This route request contains the address of the destination, along with the source node's address and a unique identification number. Each node receiving the packet checks whether it knows of a route to the destination. If it does not, it adds its own address to the route record of the packet and then forwards the packet along its outgoing links.

To limit the number of route requests propagated on the outgoing links of a node, a mobile node only forwards the route request if the request has not yet been seen by the mobile and if the mobile's address does not already appear in the route record.

A route reply is generated when the route request reaches either the destination itself, or an intermediate node that contains in its route cache an unexpired route to the destination. By the time the packet reaches either the destination or such an intermediate node, it contains a route record yielding the sequence of hops taken. Figure 2(a) illustrates the formation of the route record as the route request propagates through the network. If the node generating the route reply is the destination, it places the route

record contained in the route request into the route reply. If the responding node is an intermediate node, it appends its cached route to the route record and then generates the route reply. To return the route reply, the responding node must have a route to the initiator. If it has a route to the initiator in its route cache, it may use that route. Otherwise, if symmetric links are supported, the node may reverse the route in the route record. If symmetric links are not supported, the node may initiate its own route discovery and piggyback the route reply on the new route request. Figure 2(b) shows the transmission of route record back to the source node.

Route maintenance is accomplished through the use of route error packets and acknowledgments. Route error packets are generated at a node when the data link layer encounters a fatal transmission problem. When a route error packet is received, the hop in error is removed from the node’s route cache and all routes containing the hop are truncated at that point. In addition to route error messages, acknowledgments are used to verify the correct operation of the route links. These include passive acknowledgments, where a mobile is able to hear the next hop forwarding the packet along the route.

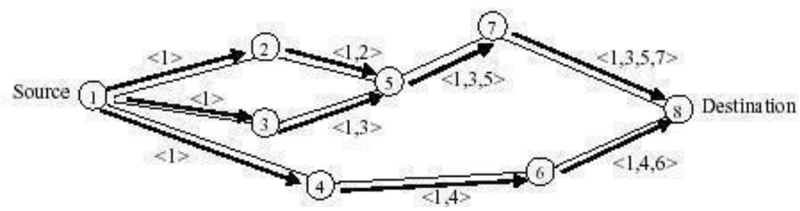


Figure 2(a) – Route discovery in DSR

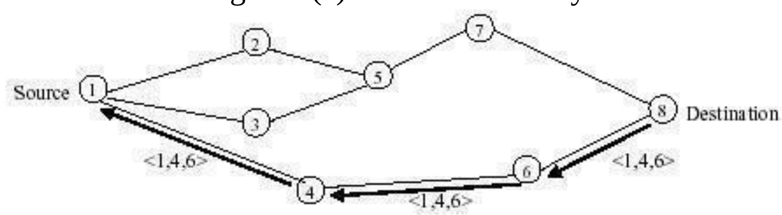


Figure 2(b) – Propagation of route reply in DSR

The Ad Hoc On-Demand Distance Vector Protocol

The Ad Hoc On-Demand Distance Vector (AODV) routing protocol [Perkins 1999] is basically a combination of DSDV and DSR. It borrows the basic on-demand mechanism of Route Discovery and Route Maintenance from DSR, plus the use of hop-by-hop routing, sequence numbers, and periodic beacons from DSDV. AODV minimizes the number of required broadcasts by creating routes on an on-demand basis, as opposed to maintaining a complete list of routes as in the DSDV algorithm. Authors of AODV classify it as a pure on-demand route acquisition system since nodes that are not on a selected path, do not maintain routing information or participate in routing table exchanges. It supports only symmetric links with two different phases:

- Route Discovery, Route Maintenance; and
- Data forwarding.

When a source node desires to send a message and does not already have a valid route to the destination, it initiates a path discovery process to locate the corresponding node. It broadcasts a route request (RREQ) packet to its neighbors, which then forwards the request to their neighbors, and so on, until either the destination or an intermediate node with a "fresh enough" route to the destination is located. Figure 3(a) illustrates the propagation of the broadcast RREQs across the network. AODV utilizes destination sequence numbers to ensure all routes are loop-free and contain the most recent route information. Each node maintains its own sequence number, as well as a broadcast ID. The broadcast ID is incremented for every RREQ the node initiates, and together with the node's IP address, uniquely identifies an RREQ. Along with the node's sequence number and the broadcast ID, the RREQ includes the most recent sequence number it has for the destination. Intermediate nodes can reply to the RREQ only if they have a route to the destination whose corresponding destination sequence number is greater than or equal to that contained in the RREQ.

During the process of forwarding the RREQ, intermediate nodes record in their route tables the address of the neighbor from which the first copy of the broadcast packet is received, thereby establishing a reverse path. If additional copies of the same RREQ are later received, these packets are discarded. Once the RREQ reaches the destination or an intermediate node with a fresh enough route, the destination/intermediate node responds by unicasting a route reply (RREP) packet back to the neighbor from which it first received the RREQ (Figure 3(b)). As the RREP is routed back along the reverse path, nodes along this path set up forward route entries in their route tables that point to the node from which the RREP came. These forward route entries indicate the active forward route. Associated with each route entry is a route timer which causes the deletion of the entry if it is not used within the specified lifetime. Because the RREP is forwarded along the path established by the RREQ, AODV only supports the use of symmetric links.

Routes are maintained as follows. If a source node moves, it is able to reinitiate the route discovery protocol to find a new route to the destination. If a node along the route moves, its upstream neighbor notices the move and propagates a link failure notification message (an RREP with infinite metric) to each of its active upstream neighbors to inform them of the breakage of that part of the route. These nodes in turn propagate the link failure notification to their upstream neighbors, and so on until the source node is reached. The source node may then choose to re-initiate route discovery for that destination if a route is still desired. An additional aspect of the protocol is the use of hello messages, periodic local broadcasts by a node to inform each mobile node of other nodes in its neighborhood. Hello messages can be used to maintain the local connectivity of a node. However, the use of hello messages may not be required at all times. Nodes listen for re-transmission of data packets to ensure that the next hop is still within reach. If such a re-transmission is not heard, the node may use one of a number of techniques, including the use of hello messages themselves, to determine whether the next hop is within its communication range. The hello messages may also list other nodes from which a mobile node has recently heard, thereby yielding greater knowledge of network connectivity.

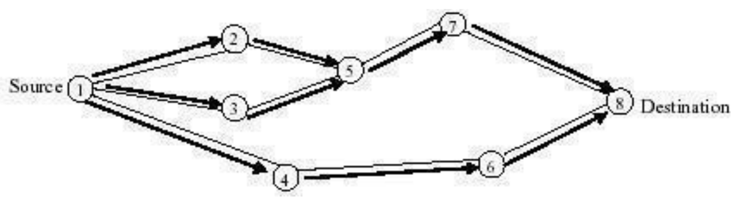
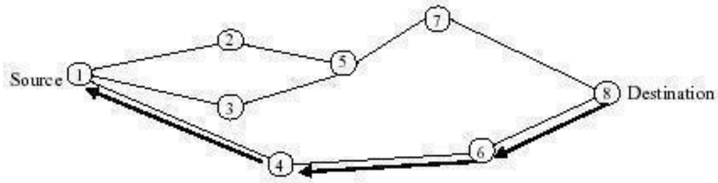


Figure 3(a) – Propagation of RREQ in AODV



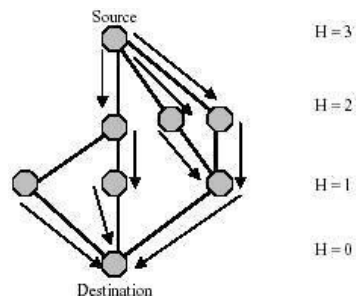


Figure 3(a) – Path taken by the RREP in AODV

Link Reversal Routing and TORA

The Temporally Ordered Routing Algorithm (TORA) [Park 1997] is a highly adaptive loop-free distributed routing algorithm based on the concept of link reversal. It is designed to minimize reaction to topological changes. A key design concept in TORA is that it decouples the generation of potentially far-reaching control messages from the rate of topological changes. Such messaging is typically localized to a very small set of nodes near the change without having to resort to a dynamic, hierarchical routing solution with its added complexity. Route optimality (shortest-path) is considered of secondary importance, and longer routes are often used if discovery of newer routes could be avoided. TORA is also characterized by a multipath routing capability.

The actions taken by TORA can be described in terms of water flowing downhill towards a destination node through a network of tubes that models the routing state of the real network. The tubes represent links between nodes in the network, the junctions of tubes represent the nodes, and the water in the tubes represents the packets flowing towards the destination. Each node has a height with respect to the destination that is computed by the routing protocol. If a tube between nodes A and B becomes blocked such that water can no longer flow through it, the height of A is set to a height greater than that of any of its remaining neighbors, such that water will now flow back out of A (and towards the other nodes that had been routing packets to the destination via A).

Figure 4 illustrates the use of the height metric. It is simply the distance from the destination node.

TORA is proposed to operate in a highly dynamic mobile networking environment. It is source initiated and provides multiple routes for any desired source/destination pair. To accomplish this, nodes need to maintain routing information about adjacent (one-hop) nodes. The protocol performs three basic functions:

- Route creation,
- Route maintenance, and
- Route erasure.

For each node in the network, a separate directed acyclic graph (DAG) is maintained for each destination. When a node needs a route to a particular destination, it broadcasts a QUERY packet containing the address of the destination for which it requires a route. This packet propagates through the network until it reaches either the destination, or an intermediate node having a route to the destination. The recipient of the QUERY then broadcasts an UPDATE packet listing its height with respect to the destination. As this packet propagates through the network, each node that receives the UPDATE sets its height to a value greater than the height of the neighbor from which the UPDATE has been received. This has the effect of creating a series of directed links from the original sender of the QUERY to the node that initially generated the UPDATE. When a node discovers that a route to a destination is no longer valid, it adjusts its height so that it is a local maximum with respect to its neighbors and transmits an UPDATE packet. If the node has no neighbors of finite height with respect to this destination, then the node instead attempts to discover a new route as described above. When a node detects a network partition, it generates a CLEAR packet that resets routing state and removes invalid routes from the network.

TORA is layered on top of IMEP, the Internet MANET Encapsulation Protocol [Corson 1997], which is required to provide reliable, in-order delivery of all routing control messages from a node to each of its neighbors, plus notification to the routing protocol whenever a link to one of its neighbors is created or broken. To reduce overhead, IMEP attempts to aggregate many TORA and IMEP control messages (which IMEP refers to as objects) together into a single packet (as an object block) before transmission. Each block carries a sequence number and a response list of other nodes from which an ACK has not yet been received, and only those nodes acknowledge the block when receiving it; IMEP retransmits each block with some period, and continues to retransmit it if needed for some maximum total period, after which TORA is notified of each broken link to unacknowledged nodes. For link status sensing and maintaining a list of a node's neighbors, each IMEP node periodically transmits a BEACON (or "BEACON-equivalent") packet, which is answered by each node hearing it with a HELLO (or "HELLO-equivalent") packet.

As we mentioned earlier, during the route creation and maintenance phases, nodes use the "height" metric to establish a DAG rooted at the destination. Thereafter,

links are assigned a direction (upstream or downstream) based on the relative height metric of neighboring nodes as shown in Figure 5(a). In times of node mobility the DAG route is broken, and route maintenance is necessary to reestablish a DAG rooted at the same destination. As shown in Figure 5(b), upon failure of the last downstream link, a node generates a new reference level that effectively coordinates a structured reaction to the failure. Links are reversed to reflect the change in adapting to the new reference level. This has the same effect as reversing the direction of one or more links when a node has no downstream links.

Timing is an important factor for TORA because the "height" metric is dependent on the logical time of a link failure; TORA assumes that all nodes have synchronized clocks (accomplished via an external time source such as the Global Positioning System). TORA's metric is a quintuple comprising five elements, namely:

- Logical time of a link failure,
- The unique ID of the node that defined the new reference level,
- A reflection indicator bit,
- A propagation ordering parameter,
- The unique ID of the node.

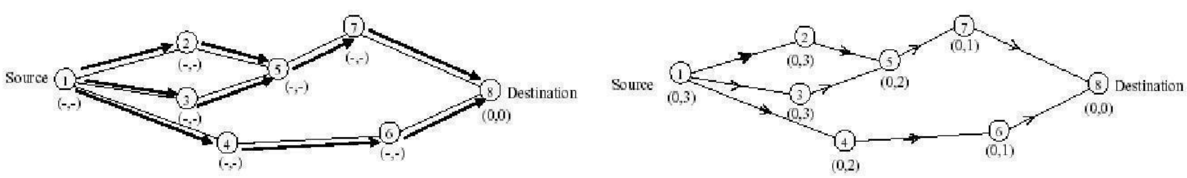


Figure 5(a) – Propagation of the query message

Figure 5(b) – Node's height updated as a result of the update message

The first three elements collectively represent the reference level. A new reference level is defined each time a node loses its last downstream link due to a link failure. TORA's route erasure phase essentially involves flooding a broadcast clear packet (CLR) throughout the network to erase invalid routes. In TORA, there is a potential for oscillations to occur, especially when multiple sets of coordinating nodes are concurrently detecting partitions, erasing routes, and building new routes based on each other (Figure 6). Because TORA uses inter-nodal coordination, its instability is similar to the "count-to-infinity" problem, except that such oscillations are temporary and route convergence ultimately occurs. Note that TORA is partially proactive and partially reactive. It is reactive in the sense that route creation is initiated on demand. However, route maintenance is done on a proactive basis such that multiple routing options are available in case of link failures.

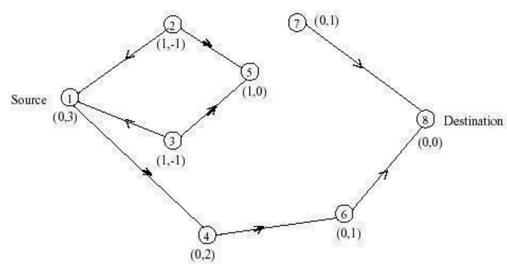


Figure 6 – Route maintenance in TORA

Hybrid Routing Protocols

Zone Routing Protocol

Zone Routing Protocol (ZRP) [Haas 1998] is a hybrid example of reactive and proactive schemes. It limits the scope of the proactive procedure only to the node's local neighborhood, while the search throughout the network, although it is global, can be performed efficiently by querying selected nodes in the network, as opposed to querying all the network nodes. In ZRP, a node proactively maintains routes to destinations within a local neighborhood, which is referred to as a routing zone and is defined as a collection of nodes whose minimum distance in hops from the node in question is no greater than a parameter referred to as zone radius. Each node maintains its zone radius and there is an overlap of neighboring zones.

The construction of a routing zone requires a node to first know who its neighbors are. A neighbor is defined as a node that can communicate directly with the node in question and is discovered through a MAC level Neighbor discovery protocol (NDP). The ZRP maintains routing zones through a proactive component called the Intrazone routing protocol (IARP) which is implemented as a modified distance vector scheme. On the other hand, the Interzone routing protocol (IERP) is responsible for acquiring routes to destinations that are located beyond the routing zone. The IERP uses a query-response mechanism to discover routes on demand.

The IERP is distinguished from the standard flooding algorithm by exploiting the structure of the routing zone, through a process known as bordercasting. The ZRP provides this service through a component called Border resolution protocol (BRP).

The network layer triggers an IERP route query when a data packet is to be sent to a destination that does not lie within its routing zone. The source generates a route query packet, which is uniquely identified by a combination of the source node's ID and request number. The query is then broadcast to all the source's peripheral nodes. Upon receipt of a route query packet, a node adds its ID to the query. The sequence of recorded node Ids specifies an accumulated route from the source to the current routing zone.If the destination does not appear in the node's routing zone, the node border casts the query to its peripheral nodes. If the destination is a member of the routing zone, a route reply is sent back to the source, along the path specified by reversing the accumulated route. A node will discard any route query packet for a query that it has previously encountered. An important feature of this route discovery process is that a single route query can return multiple route replies. The quality of these returned routes can be determined based on some metric. The best route can be selected based on the relative quality of the route.

Fisheye State Routing (FSR)

The Fisheye State Routing (FSR) protocol [Iwata 1999] introduces the notion of multi-level fisheye scope to reduce routing update overhead in large networks. Nodes exchange link state entries with their neighbors with a frequency which depends on distance to destination. From link state entries, nodes construct the topology map of the entire network and compute optimal routes. FSR tries to improve the scalability of a routing protocol by putting most effort into gathering data on the topology information that is most likely to be needed soon. Assuming that nearby changes to the network topology are those most likely to matter, FSR tries to focus its view of the network so that nearby changes are seen with the highest resolution in time and changes at distant nodes are observed with a lower resolution and less frequently. It is possible to think the FSR as blurring the sharp boundary defined in the network model used by ZRP.

Landmark Routing (LANMAR) for MANET with Group Mobility

Landmark Ad Hoc Routing (LANMAR) [Pei 2000] combines the features of FSR and Landmark routing. The key novelty is the use of landmarks for each set of nodes which move as a group (viz., a group of soldiers in a battlefield) in order to reduce routing update overhead. Like in FSR, nodes exchange link state only with their neighbors. Routes within Fisheye scope are accurate, while routes to remote groups of nodes are “summarized” by the corresponding landmarks. A packet directed to a remote destination initially aims at the Landmark; as it gets closer to destination it eventually switches to the accurate route provided by Fisheye. In the original wired landmark scheme [Tsuchiya 1988], the predefined hierarchical address of each node reflects its position within the hierarchy and helps find a route to it. Each node knows the routes to all the nodes within it hierarchical partition. Moreover, each node knows the routes to various “landmarks” at different hierarchical levels. Packet forwarding is consistent with the landmark hierarchy and the path is gradually refined from top-level hierarchy to lower levels as a packet approaches the destination.

LANMAR borrows from [Tsuchiya 1988] the notion of landmarks to keep track of logical subnets. A subnet consists of members which have a commonality of interests and are likely to move as a “group” (viz., soldiers in the battlefield, or a group of students from the same class). A “landmark” node is elected in each subnet. The routing scheme itself is modified version of FSR. The main difference is that the FSR routing table contains “all” nodes in the network, while the LANMAR routing table includes only the nodes within the scope and the landmark nodes. This feature greatly improves scalability by reducing routing table size and update traffic overhead. When a node needs to relay a packet, if the destination is within its neighbor scope, the address is found in the routing table and the packet is forwarded directly. Otherwise, the logical subnet field of the destination is searched and the packet is routed towards the landmark for that logical subnet. The packet however does not need to pass through the landmark. Rather, once the packet gets within the scope of the destination, it is routed to it directly.

The routing update exchange in LANMAR routing is similar to FSR. Each node periodically exchanges topology information with its immediate neighbors. In each update, the node sends entries within its fisheye scope. It also piggy-backs a distance vector with size equal to the number of logical subnets and thus landmark nodes.

Through this exchange process, the table entries with larger sequence numbers replace the ones with smaller sequence numbers.

FORWARDINGSTRATEGIES:

RFDR PROTOCOL

Restricted Flooding and Directional Routing (RFDR) is new proposed protocol that restricts the broadcast region will reduce routing packets, packet collisions and lowers the delay with more percentage of packet delivered.

2.1 Restricted Flooding (RF)

The main approach in restricted flooding is to restrict the flooding region. Restriction depends on distance, angle and distance covered by the next intermediate All the proposed protocols shown in figure 2 require quite complex mathematical computation of the distance, angle and coverage at all intermediate nodes to determine the nodes' participation. Information of the source and destination are required and must be inserted in the incoming packet. In MANET, route discovery is initiated by total flooding of route request (RREQ) messages that consume a large portion of the already limited bandwidth in MANET.

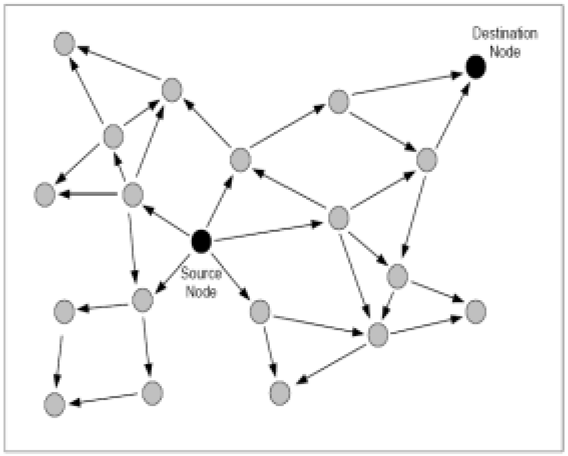


Fig. 3. Route Request (RREQs) broadcast based on Total Flooding.

As illustrated in Figure three, route request RREQ is broadcasted to all neighbours whereby frequent broadcast causes network congestion and degrades the performance of routing protocol. As we suggest utilizing restricted flooding mechanism to optimize the route establishment phase of AODVbis.

Restricted flooding is broadcasting messages to a selected number of nodes which is more than one that are located in an area in the vicinity of the destination. Location information of the destination can be obtained from any location service while location of the destination can be obtained with the aid of any other self-positioning system proposed for MANET. Then if this information is piggybacked in the reply or query packet, nodes will calculate its location with reference to the source and destination and will then decide to broadcast the query or not. Figure 4 illustrates that the same network topology .

but with restricted flooding. RREQ packets will be broadcast by nodes located in the request zone which is a quadrant drawn with respect to source node coordinates. Nodes participation is denoted by shaded circles with arrows indicating the direction of broadcast while lesser-toned circles indicate non-participating nodes. With this unique approach of using quadrant as the broadcast region, we proposed Quadrant-Based Directional Routing or RFDR.

Quadrant Based Directional Routing

QBDR is a restricted flooding routing that concentrates on a specified zone using location information provided by a location service. It restricts the broadcast region to all nodes in the same quadrant as the source and destination and does not require maintenance of a separate neighbours table at each node as in [3, 4, 5 & 6]. QBDR determines the quadrant of the current node based on the coordinates of source, destination and the current node that will direct the packet towards the destination. Even though [4] uses all these information to determine the distance or area covered, it requires trigonometric computations which will further incur delay if computed in kernel space. Decision to broadcast.

QBDR, the route request (RREQ) packet which contains the coordinates of the source and destination will be the only information the current node needs to decide to participate in the routing or not. The decision to participate at each node is made immediately as the node receives the RREQ packet and a neighbours table is not required to make the decision. RFDR will significantly reduce not only energy but also reduce the probability of packet collisions of messages rebroadcast by neighbours using the same transmission channel. This will result in

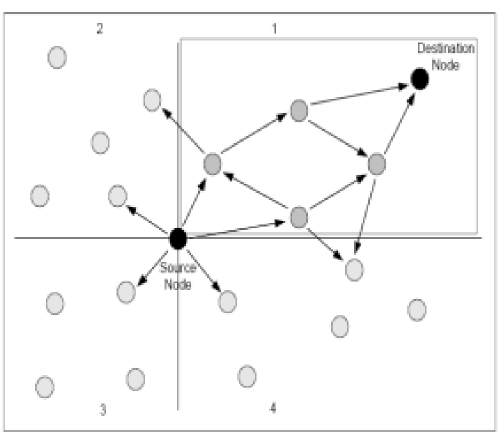


Fig.4. Route Request (RREQ) broadcast based on Restricted Flooding.

Distance Routing Effect Algorithm for Mobility

DREAM (Distance Routing Effect Algorithm for Mobility) [Basagni 1998] is a routing protocol for ad hoc networks built around two novel observations. One, called the distance effect, uses the fact that the greater the distance separating two nodes, the slower they appear to be moving with respect to each other. Accordingly, the location information in routing tables can be updated as a function of the distance separating nodes without compromising the routing accuracy. The second idea is that of triggering the sending of location updates by the moving nodes autonomously, based only on a node's mobility rate.Intuitively, it is clear that in a directional routing algorithm, routing information about the slower moving nodes needs to be updated less frequently than that about highly mobile nodes. In this way each node can optimize the frequency at which it sends updates to the networks and correspondingly reduce the bandwidth and energy used, leading to a fully distributed and self-optimizing system. Based on these routing tables, the proposed directional algorithm sends messages in the "recorded direction" of the destination node, guaranteeing delivery by following the direction with a given probability.

Routing Using Location Information

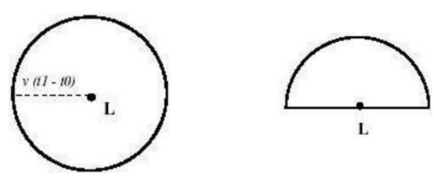
In this section we discuss some ad hoc routing protocols that take advantage of some sort of location information in the routing process.

Location-Aided Routing

The Location-Aided Routing (LAR) [Ko 1998] protocol exploits location information to limit the scope of route request flood employed in protocols such as AODV and DSR. Such location information can be obtained through GPS (Global Positioning System). LAR limits the search for a route to the so-called request zone, determined based on the expected location of the destination node at the time of route discovery. Two concepts are important to understand how LAR works: Expected Zone and Request Zone.

Let us first discuss what is an Expected Zone. Consider a node S that needs to find a route to node D. Assume that node S knows that node D was at location L at time t0, and that the current time is t1. Then, the ―expected zone‖ of node D, from the viewpoint of node S at time t1, is the region expected to contain node D. Node S can determine the expected zone based on the knowledge that node D was at location L at time t0. For instance, if node S knows that node D travels with average speed v, then S may assume that the expected zone is the circular region of radius v(t1 - t0), centered at location L (see Figure 7(a)). If actual speed happens to be larger than the average, then the destination may actually be outside the expected zone at time t1. Thus, expected zone is only an estimate made by node S to determine a region that potentially contains D at time t1.

If node S does not know a previous location of node D, then node S cannot reasonably determine the expected zone (the entire region that may potentially be occupied by the ad hoc network is assumed to be the expected zone). In this case, LAR reduces to the basic flooding algorithm. In general, having more information regarding mobility of a destination node can result in a smaller expected zone. For instance, if S knows that destination D is moving north, then the circular expected zone in Figure 7(a) can be reduced to the semi-circle of Figure 7(b).



(a) (b) Figure 7 – Examples of expected zone

Based on the expected zone, we can define the request zone. Again, consider node S that needs to determine a route to node D. The proposed LAR algorithms use flooding with one modification. Node S defines (implicitly or explicitly) a request zone for the route request. A node forwards a route request only if it belongs to the request zone (unlike the flooding algorithm in AODV and DSR). To increase the probability that the route request will reach node D, the request zone should include the expected zone (described above). Additionally, the request zone may also include other regions around the request zone.

Based on this information, the source node S can thus determine the four corners of the expected zone. S includes their coordinates with the route request message transmitted when initiating route discovery. When a node receives a route request, it discards the request if the node is not within the rectangle specified by the four corners included in the route request. For instance, in Figure 8, if node I receives the route request from another node, node I forwards the request to its neighbors, because I determines that it is within the rectangular request zone. However, when node J receives the route request, node J discards the request, as node J is not within the request zone (see Figure 8).

The algorithm just described in the called LAR scheme 1. The LAR scheme 2 is a slight modification to include two pieces of information within the route request packet: assume that node S knows the location (Xd; Yd) of node D at some time t0 – the time at which route discovery is initiated by node S is t1, where t1 ≥ t0. Node S calculates its distance from location (Xd; Yd), denoted as DISTS, and includes this distance with the route request message. The coordinates (Xd; Yd) are also included in the route request packet. With this information, a given node J forwards a route request forwarded by I (originated by node S), if J is within an expected distance from (Xd; Yd) than node I.

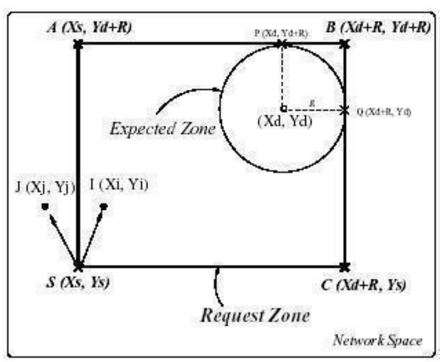


Figure 8 – LAR scheme .

Comparison Table

Table 2 summarizes the main characteristics of the most cited protocols discussed so far.

Table 2 – Protocol characteristics

Routing Protocol	Route Acquisition	Flood for Route Discovery	Delay for Route Discovery	Multipath Capability	Upon Route Failure
DSDV	Computed a priori	No	No	No	Floods route updates throughout the network
WRP	Computed a priori	No	No	No	Ultimately, updates the routing tables of all nodes by exchanging MRL between neighbors
DSR	On-demand, only when needed	Yes. Aggressive use of caching often reduces flood scope	Yes	Not explicitly. The technique of salvaging may quickly restore a Route	Route error propagated up to the source to erase invalid path
AODV	On-demand, only when needed	Yes. Conservative use of cache to reduce flood scope	Yes	No, although recent research indicate viability	Route error broadcasted to erase invalid path
TORA	On-demand, only when needed	Usually, only one flood for initial DAG construction	Yes. Once the DAG is constructed, multiple paths are found	Yes	Error is recovered locally, and only when alternative routes are not available
LAR	On-demand, only when needed	Localized flood by using location information	Yes	No	Route error propagated up to the source
ZRP	Hybrid	Only outside a source's zone	Only if the destination is outside the source's zone	No	Hybrid of updating nodes' tables within a zone and propagating route error to the source